

Task 3: Wireshark Packet Analysis

Task Explanation

Wireshark is a network protocol analyzer used to capture and inspect network traffic. In this task, Wireshark was used in the TryHackMe AttackBox to capture traffic generated while interacting with the DVWA application through the ens5 interface. HTTP, DNS, and TCP traffic were analyzed, including packet details, IP addresses, port numbers, and TCP flags. The captured traffic was also checked for unusual or malformed packets.

Tools Used

- TryHackMe
- Wireshark
- DVWA
- Firefox Browser
- Linux AttackBox

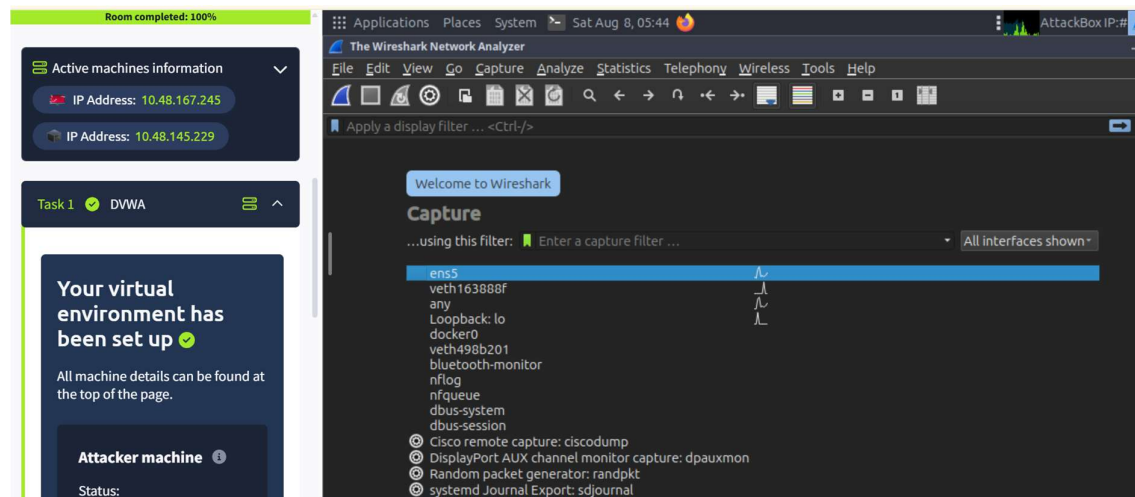
3. Procedure

Step 1: Open Wireshark

Wireshark was opened from the AttackBox.

The **ens5** network interface was selected to capture network traffic.

Screenshot 1: Wireshark with the ens5 interface selected.

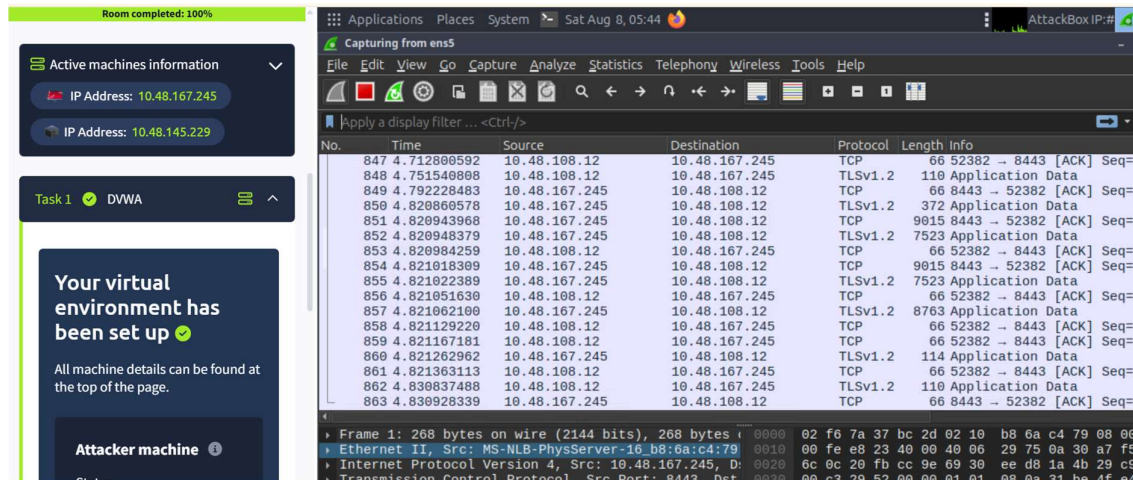


Step 2: Capture Network Traffic

The packet capture was started by double-clicking the **ens5** interface.

Traffic was generated by accessing the DVWA application through Firefox.

The captured packets appeared in the Wireshark packet list.



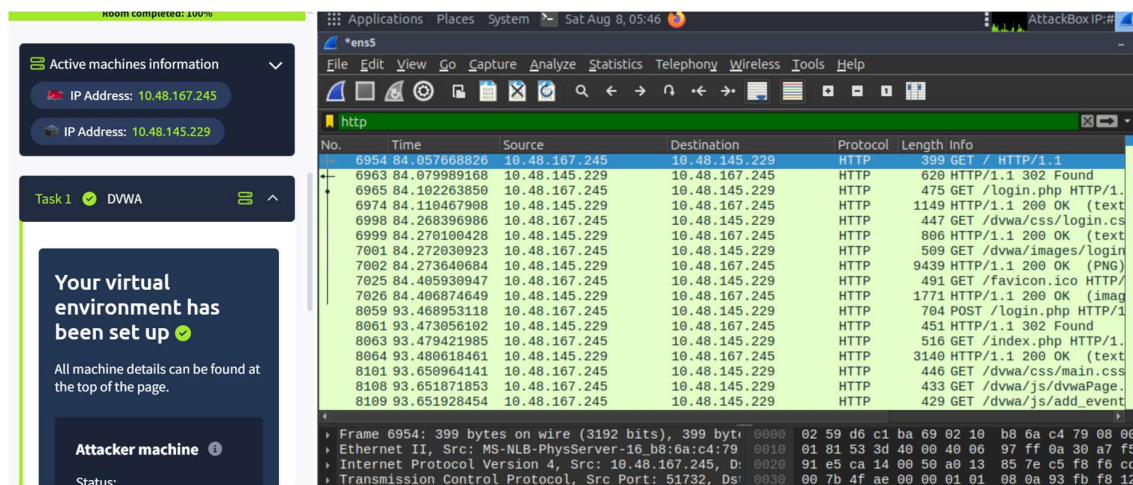
Step 3: Analyze HTTP Traffic

The following Wireshark display filter was used:

http

This filter displays HTTP packets captured during communication with the web application.

The packets were examined to identify the source and destination addresses, ports, and HTTP communication.



Step 4: Analyze DNS Traffic

The following display filter was used:

dns

This filter displays DNS query and response packets.

The DNS packets were examined to understand how domain-name resolution takes place.

Active machines information

IP Address: 10.48.167.245

IP Address: 10.48.145.229

Task 1 DVWA

Your virtual environment has been set up

All machine details can be found at the top of the page.

Attacker machine

Status:

Step 5: Analyze TCP Handshake

TCP traffic was filtered using:

tcp

TCP packets involved in establishing a connection were examined.

The TCP three-way handshake consists of:

SYN → SYN-ACK → ACK

The TCP flags were inspected to understand the connection establishment process.

Room completed: 100%

Active machines information

IP Address: 10.48.167.245

IP Address: 10.48.145.229

Task 1 DVWA

Your virtual environment has been set up

All machine details can be found at the top of the page.

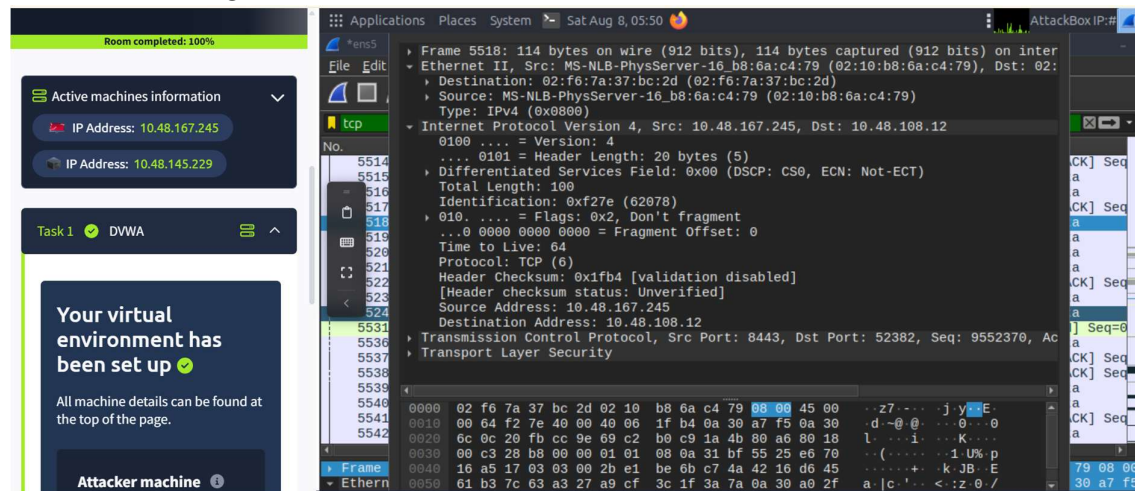
Attacker machine

Step 6: Analyze Packet Structure and Flags

A TCP packet was expanded in Wireshark to examine its detailed structure.

The following information was analyzed:

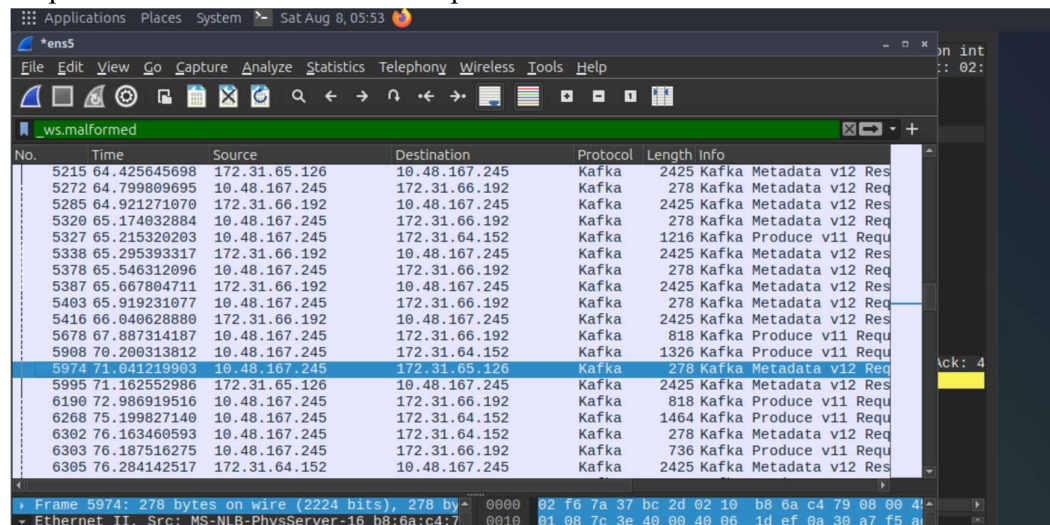
- Source IP address
- Destination IP address
- Source port
- Destination port
- Sequence number
- Acknowledgment number
- TCP flags
- Packet length

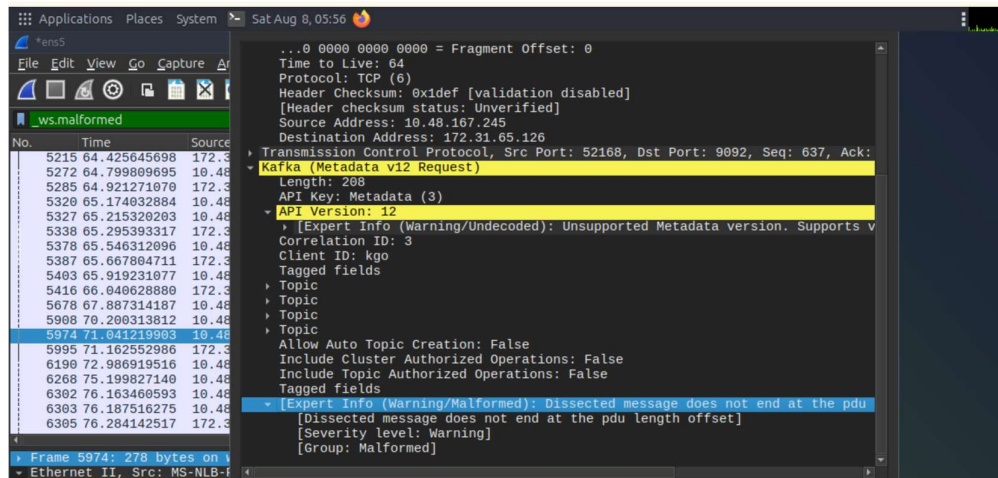


Step 7: Check for Unusual or Malformed Packets

The captured packets were reviewed for unusual traffic patterns, abnormal TCP flags, retransmissions, or malformed packets.

The packet information and protocol details were examined to determine whether any suspicious or malformed traffic was present.





4. Findings

- HTTP traffic was successfully captured and identified.
- DNS query and response packets were identified.
- TCP traffic and the three-way handshake were analyzed.
- TCP packet structure and flags were examined.
- Wireshark identified packets matching the `_ws.malformed` filter.
- The malformed packets were reviewed as part of the network traffic analysis.

5. Security Risks

The analysis of network traffic can help identify several potential security risks:

- Unencrypted HTTP traffic may expose information transmitted between the browser and server.
- Unusual TCP traffic may indicate connection problems or suspicious activity.
- Malformed packets may result from network errors, capture issues, unusual protocol behavior, or potentially malicious traffic.
- Unexpected network communication may indicate unauthorized activity.

6. Analysis

- Wireshark provided detailed information about the network packets exchanged during communication with the testing environment.
- HTTP packets were analyzed to identify web traffic, while DNS packets demonstrated domain-name resolution. TCP packets were examined to understand the three-way handshake and the purpose of TCP flags such as SYN and ACK.
- The `_ws.malformed` filter was also used to identify packets that Wireshark considered malformed. These packets were reviewed as part of the security analysis.

7. Result

Network traffic was successfully captured and analyzed using Wireshark. HTTP, DNS, and TCP traffic were identified, the TCP handshake and packet structure were examined, and packets flagged as malformed were investigated using Wireshark filters.